θhetaGate

ONE-PAGE WRITE-UP · ALPACA AI TRADING AGENTS HACKATHON

ThetaGate is an autonomous **paper-only** options desk. It sells defined-risk credit spreads when at-the-money implied volatility is rich versus 20-day realized volatility, and stands down in cheap volatility or a simple breakout. It never takes unlimited short premium and never lets the model place an order.

Track: **Options Alpha Agents** Mode: **paper only** Paper account ID: **PA3KII0I2OJ1**

AI LOGIC

In the premarket window the desk builds a persisted 10-to-20-name watchlist from Alpaca market data (most-actives, movers, recent news), not a hardcoded SPY / QQQ / IWM list, and refreshes it once after midday ET; a failed refresh keeps the verified morning list. Warrants, OTC issues, and levered products are excluded. Names under $10 are dropped when a price is available; if the whole price request fails, that filter is skipped for the cycle. A failed initial discovery yields an empty book, never a silent fallback.

Each cycle, Python fetches bars and a DTE-banded chain, derives missing IV and delta with Black-Scholes, and gates on IV / RV ≥ 1.2. Only surviving short and long candidates reach the model. Featherless proposes a two-leg `TradeProposal`; malformed JSON is retried (≤ 3) and, if `XAI_FALLBACK` is on, handed to Grok. A short critic can challenge the idea but cannot approve or spend. The model never sees `place_option_order`.

RISK GATES

Python independently recomputes max loss `(width − credit) × 100 × qty` from mid quotes and rejects anything above 2% of NAV. It also enforces DTE 7 to 21, short delta 0.20 to 0.30, long delta 0.10 to 0.15, bid-ask width, IV sanity, book caps (max 3 structures, 2 per underlying), daily −3% and total −8% equity halts, a cooldown, and a file kill switch (`logs/KILL`). Invalid proposals become vetoes.

Entry is a day limit at the **natural** credit (short bid − long ask) so an approved ticket can fill. Because that credit is below the mid, the filled structure's defined risk runs about half of each leg's bid-ask spread above the mid-based estimate, so the 2% cap is enforced on the mid figure rather than the fill. `PENDING_ENTRY` becomes `OPEN` only after Alpaca reports `filled_qty > 0`. A halt cancels unfilled entries, reconciles, then flattens live exposure, and stays incomplete until both pending and open risk are gone. Closes are an atomic debit sized to the reconciled `open_qty`, not a copied opening credit.

ALPACA INFRASTRUCTURE

The official MCP server (`alpaca-mcp-server`) is the broker interface for account, clock, bars, chain, news, and multi-leg `place_option_order` (`qty` plus OCC `symbol` / `ratio_qty`, `limit_price` negative for a credit); it satisfies the event's MCP-or-CLI requirement. Market-data REST drives the screener and snapshots. The repo also wires one read-only `alpaca account` CLI call. Paper only: `ALPACA_PAPER_TRADE=false` is rejected. Submit is fail-closed on `EXPECTED_ACCOUNT_ID` versus the live `account_number`. Railway runs the Streamlit desk and the `--live` scheduler together; the desk is monitor plus STOP, not the trader.

EVIDENCE

Across three paper sessions on account `PA3KII0I2OJ1`, ThetaGate completed 372 cycles, stood down 81% of the time, filled nine entries, closed seven round trips, and finished at $99,929.76 with no unresolved lifecycle states.

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github.com/theaimlessfox-hackathons/alpaca-ai-trading-agents